

Session 04 (AIML) -Assignment Question

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- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
#session 4
#Q1
# Lambda Function
square = lambda x: x * x

num = int(input("Enter a number: "))
print("Square using lambda:", square(num))

# Normal Function
def square_def(x):
    return x * x

print("Square using def:", square_def(num))

# Comparison:
# lambda -> used for short, single-line functions.
# def -> used for larger functions with multiple statements.
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Pyt
-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '55498' '--' 'c:
Enter a number: 22
Squ
Squ Focus folder in explorer (ctrl + click)
PS C:\Users\tanve\Desktop\ITR> □
```

Q2.

PROGRAM:

```
session 4.py / ...
1  #Q2
2
3  # Lambda function to add three numbers
4  add_three = lambda a, b, c: a + b + c
5
6  a = int(input("Enter first number: "))
7  b = int(input("Enter second number: "))
8  c = int(input("Enter third number: "))
9
10 print("Sum =", add_three(a, b, c))45
1
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\
.vscode\extensions\ms-python.debugpy-2026.6.0-win32-
Enter first number: 45
Enter second number: 5
Enter third number: 5
Sum = 55
PS C:\Users\tanve\Desktop\ITR> |
```

Q3.

PROGRAM:

```
session 4.py > ...
1  #Q3
2
3  # a) Print numbers from 1 to 25
4  print("Numbers from 1 to 25:")
5  for i in range(1, 26):
6      print(i)
7
8  # b) Sum of numbers from 1 to 20
9  total = 0
10 for i in range(1, 21):
11     total += i
12
13 print("Sum from 1 to 20 =", total)
14
15 # c) Table of 5
16 print("Table of 5:")
17 for i in range(1, 11):
18     print(f"5 x {i} = {5 * i}")
19
20 # For loop is used when the number of iterations is known.
```

OUTPUT:

Numbers from 1 to 25:

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25

Sum from 1 to 20 = 210

Table of 5:

$5 \times 1 = 5$
 $5 \times 2 = 10$
 $5 \times 3 = 15$
 $5 \times 4 = 20$
 $5 \times 5 = 25$
 $5 \times 6 = 30$
 $5 \times 7 = 35$
 $5 \times 8 = 40$
 $5 \times 9 = 45$
 $5 \times 10 = 50$

PS: C:\Users\tanve\Desktop\ITP>

Q4.

PROGRAM:

```
session 4.py > ...
1  #Q4
2
3  total = 0
4  positive number (0 or negative to stop): ")
5
6      if num <= 0:
7          break
8
9      total += num
10
11 print("Total Sum =", total)
```

OUTPUT:

```
.\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x
Enter a positive number (0 or negative to stop): 4
Enter a positive number (0 or negative to stop): 0
Total Sum = 37
PS C:\Users\tanve\Desktop\ITR> █
ed.
```

Q5.

PROGRAM:

```
session 4.py > ...
1  #Q5
2
3  import math
4
5  num = float(input("Enter a number: "))
6
7  # Square Root
8  print("Square Root =", math.sqrt(num))
9
10 # Factorial (only for positive integers)
11 if num >= 0 and num.is_integer():
12     print("Factorial =", math.factorial(int(num)))
13 else:
14     print("Factorial not possible for this number.")
15
16 # Ceiling Value
17 print("Ceiling Value =", math.ceil(num))
18
19 # Floor Value
20 print("Floor Value =", math.floor(num))
21
22 # Math module provides useful mathematical functions
23 # such as sqrt(), factorial(), ceil(), floor(), etc.11
24
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Use
.vscode\extensions\ms-python.debugpy-2026.6.0
Factorial = 39916800
Ceiling Value = 11
Floor Value = 11
PS C:\Users\tanve\Desktop\ITR> █
```

Q6.

PROGRAM:

```
session 4.py > ...
1  #Q6
2
3  import random
4
5  # 1. Generate and print 5 random integers between 1 and 100
6  print("5 Random Integers:")
7  for i in range(5):
8      print(random.randint(1, 100))
9
10 # 2. Generate and print one random number between 50 and 150
11 print("\nRandom Number between 50 and 150:")
12 print(random.randrange(50, 151))
13
14 # 3. Print a random floating point number between 0 and 1
15 print("\nRandom Float between 0 and 1:")
16 print(random.random())
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Users\tanve\
.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher
5 Random Integers:
58
85
21
10
59

Random Number between 50 and 150:
135

Random Float between 0 and 1:
0.03250846569614063
PS C:\Users\tanve\Desktop\ITR> █
```

Q7.

PROGRAM:

```
session 4.py
1   #Q7
2
3   # 1. Import entire math module
4   import math
5   print("2^4 using math.pow():", math.pow(2, 4))
6
7   # 2. Import only sqrt
8   from math import sqrt
9   print("Square root of 25:", sqrt(25))
10
11  # 3. Import math with alias
12  import math as m
13  print("Factorial of 5:", m.factorial(5))
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR'
2^4 using math.pow(): 16.0
Square root of 25: 5.0
Factorial of 5: 120
PS C:\Users\tanve\Desktop\ITR>
```


Q8.

PROGRAM:

```
mymodule.py > calculate_power
1  # Custom module
2
3  def greet(name):
4      print("Hello", name + ", Welcome to Python Class!")
5
6  def calculate_power(base, exp):
7      return base ** exp
```

```
Import custom module.py > ...
1  # Import custom module
2
3  import mymodule
4
5  name = input("Enter your name: ")
6  mymodule.greet(name)
7
8  base = int(input("Enter base: "))
9  exp = int(input("Enter exponent: "))
10
11 result = mymodule.calculate_power(base, exp)
12
13 print("Power =", result)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:: cd 'c:\Users\tanve\Desktop\
.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle

Enter your name: shruti darwatkar
Hello shruti darwatkar, Welcome to Python Class!
Enter base: 8
Enter exponent: 5
Power = 32768
PS C:\Users\tanve\Desktop\ITR> 
```

Q9.

PROGRAM:

```
mymodule.py > ...
1  #Q9
2
3  def show_message():
4      print("This is a function.")
5
6  if __name__ == "__main__":
7      print("Program is running directly.")
8      show_message()
```

OUTPUT:

```
● PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\AppData\Local\Microsoft\Windows\Inetput\ms-python.debugpy-2026.6.0-1'
Program is running directly.
This is a function.
○ PS C:\Users\tanve\Desktop\ITR> █
```

Q10.

PROGRAM:

```
mymodule.py > ...
1  import math
2  import random
3  # Lambda Function
4  square = lambda x: x * x
5  # Normal Function
6  def calculate_power(base, exp):
7      return math.pow(base, exp)
8
9  while True:
10     print("\n--- Simple Math Utility ---")
11     print("1. Square")
12     print("2. Power")
13     print("3. Random Number")
14     print("4. Exit")
15
16     choice = input("Enter your choice: ")
17
18     if choice == "1":
19         num = int(input("Enter a number: "))
20         print("Square =", square(num))
21
22     elif choice == "2":
23         base = int(input("Enter base: "))
24         exp = int(input("Enter exponent: "))
25         print("Power =", calculate_power(base, exp))
26
27     elif choice == "3":
28         print("Random Number =", random.randint(1, 100))
29
30     elif choice == "4":
31         print("Program Ended.")
32         break
33
34     else:
35         print("Invalid Choice! Please try again.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Users\tanve\hon.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '62133' '--' 'c:\Users\ta
1. Square
2. Power
3. Random Number
4. Exit
Enter your choice: 2
Enter base: 4
Enter exponent: 3
Power = 64.0

--- Simple Math Utility ---
1. Square
2. Power
3. Random Number
4. Exit
Enter your choice: 3
Random Number = 99

--- Simple Math Utility ---
1. Square
2. Power
3. Random Number
4. Exit
Enter your choice: 3
Random Number = 60

--- Simple Math Utility ---
1. Square
2. Power
3. Random Number
4. Exit
Enter your choice: 80
Invalid Choice! Please try again.

--- Simple Math Utility ---
1. Square
2. Power
3. Random Number
4. Exit
Enter your choice: 4
Program Ended.
PS C:\Users\tanve\Desktop\ITR> █
```